MAHADEER A

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Academic Profile

Doctoral researcher in Mathematics specializing in Petri net theory, formal language theory, image encryption, and image segmentation. Seeking an Assistant Professor position to contribute through effective mathematics teaching, interdisciplinary research, student mentoring, and academic service.

Education

- **Ph.D. in Mathematics (Thesis Submitted)** — *Dec. 2022–Present*
  SRM Institute of Science and Technology — Kattankulathur, Tamil Nadu
  - **Thesis:** *A Study on Petri Net Models for Partial Languages, Image Segmentation, and Image Encryption*
- **M.Sc. Mathematics** — *2022*
  The New College, University of Madras — Chennai, Tamil Nadu
  - **CGPA:** 9.084; First Class with Exemplary; **Gold Medalist**
- **B.Sc. Mathematics** — *2020*
  The New College, University of Madras — Chennai, Tamil Nadu
  - **CGPA:** 8.566; First Class with Distinction
- **Higher Secondary Certificate (HSC)** — *2017*
  MEASI Matriculation Higher Secondary School — Chennai, Tamil Nadu
  - **Percentage:** 83.16%
- **Secondary School Leaving Certificate (SSLC)** — *2015*
  MEASI Matriculation Higher Secondary School — Chennai, Tamil Nadu
  - **Percentage:** 93.20%

Research Experience

- **Doctoral Researcher** — *Dec. 2022–Present*
  Department of Mathematics, SRM Institute of Science and Technology — Kattankulathur, Tamil Nadu
  - Conducted theoretical research on Petri nets for generating partial and Watson–Crick partial languages, with emphasis on language classes and closure properties.
  - Developed Petri-net-based approaches for image segmentation and image encryption, with the encryption frameworks incorporating cryptographic hashing and chaotic maps.
  - Published research articles in high-impact, peer-reviewed journals.
  - Collaborated with researchers on interdisciplinary studies leading to peer-reviewed publications.

Research Publication Summary

Total Research Publications:	08
Published Journal Articles:	07
Book Chapters:	01
International Journals:	*International Journal of Information Technology* (2); *IET Image Processing* (1); *New Mathematics and Natural Computation* (1); *Information Sciences* (1); *Nonlinear Dynamics* (1); and *Journal of Cybersecurity* (1).
Cumulative Impact Factor:	19.6
H-index:	2

Publications

S.No.	Publications	Indexing & Metrics
1	**Mahadeer A**, Arulprakasam R, Gurusamy R, and Yilun Shang. "Enhancing Grayscale Image Encryption through Petri Net Dynamics and Hash-Extracted Keystreams," *Journal of Cybersecurity*, **12**(1) (2026), 1–19. https://doi.org/10.1093/cybsec/tyag019.	ESCI, Scopus, IF: 4.4, Q1
2	**Mahadeer A** and Arulprakasam R. "Petri Net-Driven Expanding Tent–Sine Chaotic Keystream Generator with Adaptive Token Flow for Image Encryption," *Information Sciences*, **748** (2026), 123515. https://doi.org/10.1016/j.ins.2026.123515.	SCIE, Scopus, IF: 6.0, Q1
3	**Mahadeer A** and Arulprakasam R. "Image Encryption Using Petri Net Dynamics with Pixel Permutation and Keystream Masking," *Nonlinear Dynamics*, **114**(8) (2026), 547. https://doi.org/10.1007/s11071-026-12413-y.	SCIE, Scopus, IF: 5.7, Q1
4	**Mahadeer A**, Arulprakasam R, Gurusamy R, and Yilun Shang. "A Petri Net Token-Flow PRNG for Grayscale Image Encryption," *New Mathematics and Natural Computation*, 2026. https://doi.org/10.1142/S179300572850041X.	ESCI, Scopus, IF: 1.1, Q3

S.No.	Publications	Indexing & Metrics
5	**Mahadeer A**, Arulprakasam R, Gurusamy R, and Yilun Shang. "Adaptive Petri Net Token Flow Approach for Enhanced Grayscale Image Segmentation," *IET Image Processing*, **19**(1) (2025), e70255. https://doi.org/10.1049/ipr2.70255.	SCIE, Scopus, IF: 2.4, Q2
6	**Mahadeer A**, Arulprakasam R, and Dare V R. "Watson–Crick Partial Petri Net Languages and their Properties," *International Journal of Information Technology*, **17**(4) (2025), 2553–2569. https://doi.org/10.1007/s41870-024-02272-4.	Scopus, SNIP: 1.637, Q1
7	**Mahadeer A**, Arulprakasam R, and Dare V R. "Partial Petri Net Languages and their Properties," *International Journal of Information Technology*, **16**(6) (2024), 3663–3676. https://doi.org/10.1007/s41870-024-01903-0.	Scopus, SNIP: 1.637, Q1

Book Chapter

S.No.	Book Chapter	Publisher & Year
1	**Mahadeer A** and Arulprakasam R. "A Partial Petri Net Framework for Addressing Uncertainty in Healthcare Processes," in *Applied Mathematics, Automation and Computing*. https://doi.org/10.1201/9781003770497.	CRC Press, Taylor & Francis, 2026

Patent

S.No.	Patent	Application Details
1	**Mahadeer A**, Arulprakasam R, and Dhanalakshmi S. "Petri Net-Based Image Encryption Using Pixel Permutation and Keystream Masking."	Indian Patent Application No. 202641016722 A, 2026

Peer-Reviewer

Serve as an ad hoc reviewer for the following international peer-reviewed journals:

S.No.	Journal	Publisher
1	*Information Sciences*	Elsevier
2	*Information Processing & Management*	Elsevier
3	*Chaos, Solitons & Fractals*	Elsevier
4	*Journal of Information Security and Applications*	Elsevier
5	*Scientific Reports*	Nature Portfolio

Conferences

- Presented the paper entitled **"A Partial Petri Net Framework for Addressing Uncertainty in Healthcare Processes"** at the *CSIR & TNSCST-Sponsored International Conference on Applied Mathematics, Automation and Computing (ICAMAC 2025)*, organized by Kongu Engineering College, Perundurai, Tamil Nadu, India, February 21–22, 2025.
- Presented the paper entitled **"Modeling Uncertainty in Healthcare Process Management with Partial Petri Nets"** at the *12th National Conference on Mathematical Techniques and Applications (NCMTA-2025)*, organized by the Department of Mathematics, SRM Institute of Science and Technology, Kattankulathur, January 9–10, 2025.
- Presented the paper entitled **"Properties of Partial Petri Net Languages"** at the *Dr. Paarivendhar Research Colloquium (DPRC-2024)*, SRM Institute of Science and Technology, Kattankulathur, March 26–28, 2024.
- Presented the paper entitled **"Partial Petri Net Languages and their Properties"** at the *International Conference on Emerging Frontiers in Nonlinear Complex Systems, Computational Intelligence and their Applications (ICNCS 2024)*, organized by Vellore Institute of Technology, Chennai, February 7–9, 2024.

Workshops and Professional Development

- **Generative AI and Prompt Engineering** — Five-Day Virtual Faculty Development Programme organized by the Department of CSE (Emerging Technologies), SRM Institute of Science and Technology, Vadapalani, November 25–29, 2024.
- **Recent Frontiers in Pure & Applied Mathematics** — Five-Day Virtual Faculty Development Programme organized by the Department of Mathematics, Kongunadu Arts and Science College (Autonomous), under the DBT Star College Scheme, Coimbatore, November 11–15, 2024.
- **Research Proposal Writing for Funding Projects: Opportunities and Future Aspects** — International Workshop organized by Inspire Softech Solutions, Chennai, September 20–22, 2024.
- **Mathematics Behind Artificial Intelligence & Machine Learning** — Three-Day Workshop organized by the Department of Mathematics, SRM Institute of Science and Technology, Kattankulathur, May 16–18, 2024.
- **Mathematical Modelling and Its Applications** — International Workshop organized by the Division of Mathematics, School of Advanced Sciences, Vellore Institute of Technology, Chennai, December 13–14, 2023.
- **Advances in Graph Theory** — Five-Day Online Faculty Development Programme organized by the Division of Mathematics, School of Advanced Sciences, Vellore Institute of Technology, Chennai, September 11–15, 2023.
- **Role of Mathematics in Machine Learning** — Five-Day Faculty Development Programme for Skill Development organized by the Department of Mathematics, School of Basic Sciences, SRM Institute of Science and Technology, Kattankulathur, February 20–24, 2023.

- **Numerical and Analytical Techniques in Engineering Problems (IWNATEP-2023)** — Third International Workshop organized by the Department of Mathematics, SRM Institute of Science and Technology, Kattankulathur, January 30–February 1, 2023.

NPTEL Certifications

S.No.	Course Details	Certification & Score
1	**Introduction to Automata, Languages and Computation** IIT Kharagpur (12-week course; January–April 2023)	**Elite + Gold**; Score: 91%
2	**Graph Theory** IISER Pune (8-week course; January–March 2022)	**Elite + Silver**; Score: 77%
3	**Introduction to Abstract and Linear Algebra** IIT Kharagpur (8-week course; August–October 2021)	**Elite**; Score: 73%
4	**Introduction to Probability and Statistics** IIT Madras (4-week course; January–February 2021)	**Elite**; Score: 61%

Technical Skills

- **Document Preparation:** LaTeX (advanced proficiency)
- **Office Applications:** Microsoft Word, PowerPoint, and Excel (proficient)
- **Programming:** Python (working knowledge)

Honors and Awards

- **Dr. A.P.J. Abdul Kalam Research Award 2026** *February 2026*
  SRM Institute of Science and Technology
  - Conferred for excellence in doctoral research, based on a first-authored publication indexed in Web of Science/Scopus during the calender year 2025.

Leadership Experience

- **Student Secretary, Department of Mathematics** *2019–2020*
  The New College, University of Madras Chennai, Tamil Nadu
  - Coordinated departmental student activities and served as a liaison between students and faculty members.

Additional Information

Gender: Male

Languages Known: English (speak, read, and write); Tamil (speak, read, and write); Arabic (read and write)

References

1. **Dr. R. Arulprakasam**
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2. **Dr. M. Rajesh Kannan**
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3. **Dr. Sairam Kaliraj**
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Declaration

I hereby declare that the information provided above is true and correct to the best of my knowledge.

MAHADEER A